Experiment No:7

Understanding dispersion hardening and effect of particulate concentration on microstructure and mechanical properties

- **Objectives:**

  a. Prepare castings with different particulate fraction (5% & 20% of SiC in Al)
  b. Understand effect of particulate concentration on dispersion hardening
  c. Study the influence of particulate fraction on microstructure of the material

- **Experimental Set-up:**

  Arc melter for casting, abrasive cut-off machine, hot-mounting press, for sample preparation. Belt grinder and double disc polishing machine for sample polishing.

  Finally sample are etched using recommended etchant and microstructure is studied using metallurgical microscope. Hardness of the casted alloy measured using Vickers' Hardness Tester.

- **Consumable required:**
  - Aluminium foil and silicon carbide powder, abrasive paper of different grit sizes, and chemicals.
- **Theory:**
  - Metals and metal alloys may be strengthened and hardened by the uniform dispersion of fine particles of a very hard and inert material. The dispersed phase may be metallic or non-metallic.
  - The strengthening mechanism involves interactions between the particles and dislocations within the matrix.
  - Important aspect of the strengthening processes is that they operate by restricting or stopping the movement of dislocations and, therefore, the applied stress needed to induce plastic flow increases resulting in a corresponding increase in strength.
  - The dispersion strengthening depends upon the type and size of the particulate used. Further, fraction of the particulate material used also plays an important role. Closer the particles are to each other; more difficult it becomes for the dislocations to bypass the particles. Simultaneously, for the same fraction used, small size particles are more effective in pinning the dislocation motion.

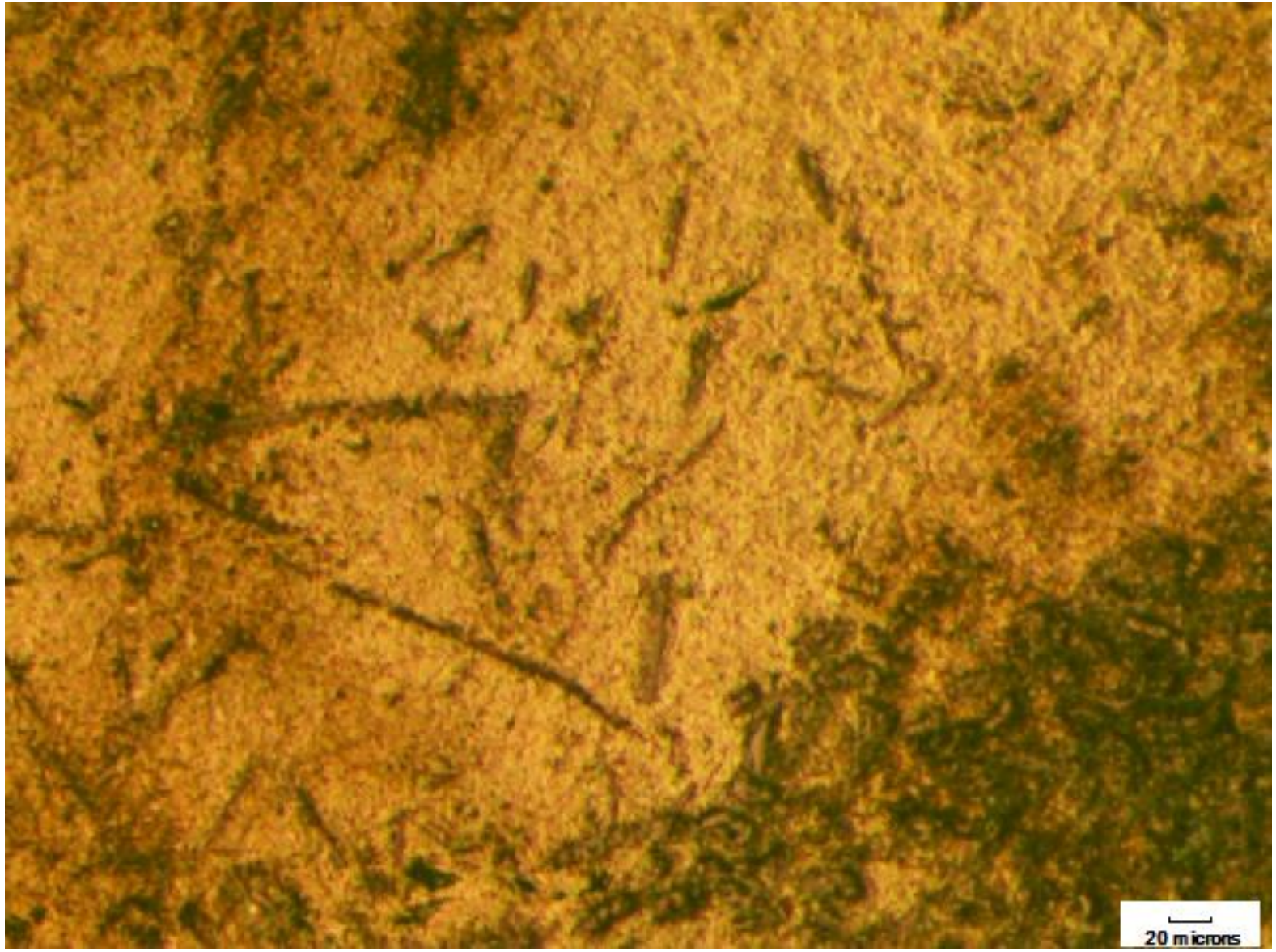

Fig: 5% SiC in Al

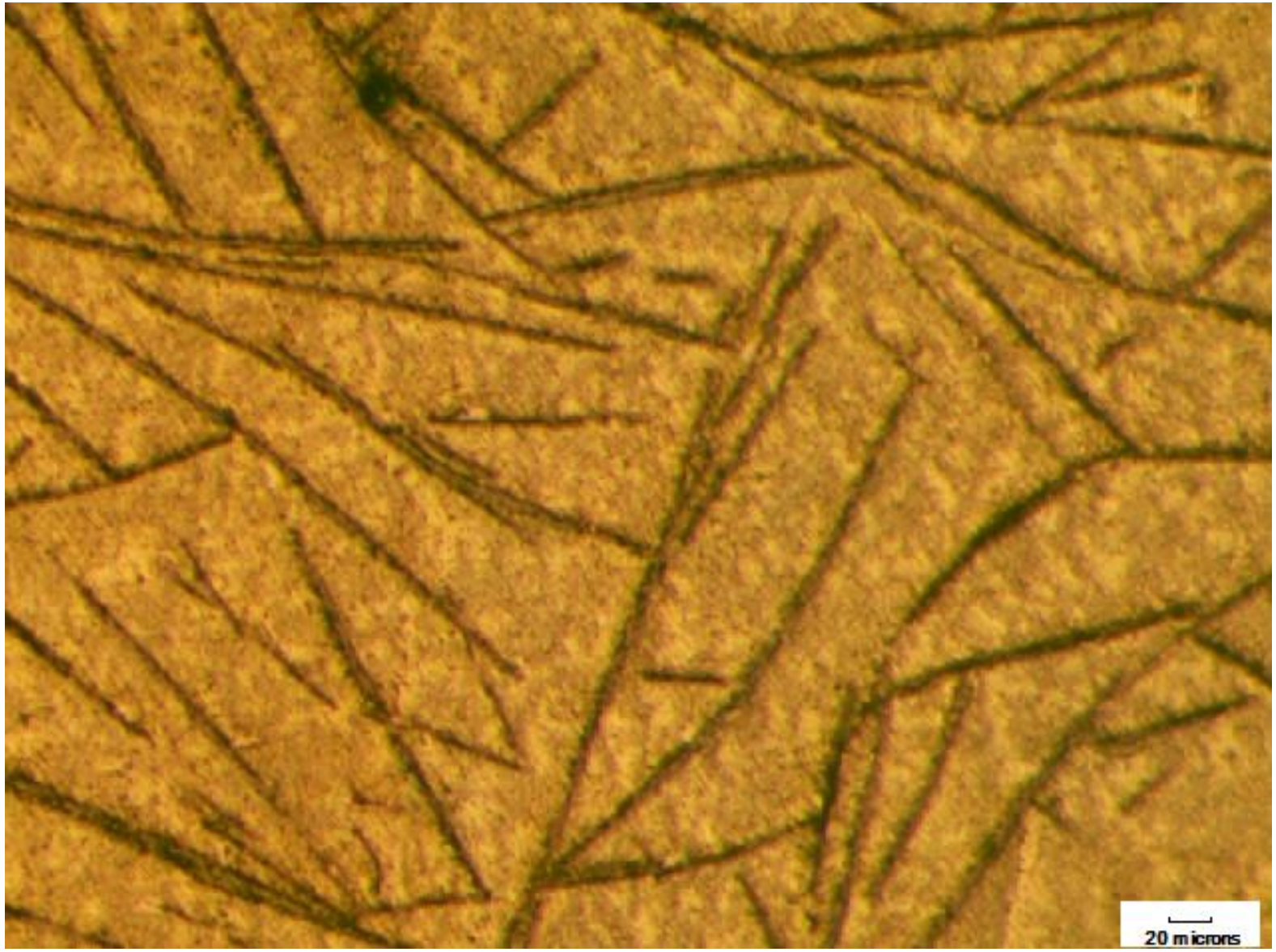

Fig: 20% SiC in Al